CTO, Software Architect, and Framework-Less Engineering Leader

Language-agnostic systems design | Architecture-first execution

Seasoned **software architect** with 15+ years of experience designing robust systems across financial services, SaaS products, and AI-driven platforms. As **CTO at Ultrai**, I lead architecture, engineering delivery, hiring, infrastructure, and security while remaining hands-on in implementation and technical spikes. I bridge **DevOps**, databases, and system design while prioritizing craftsmanship and elegant solutions in every project.

I focus on **framework-less engineering** where possible, using strong design principles, clear boundaries, and stateless patterns to keep systems maintainable over time. My career demonstrates consistent adaptation across stacks and domains, selecting each language pragmatically to maximize reliability, maintainability, and delivery speed.

Beyond implementation, I **mentor engineers** in Python best practices and functional programming principles. My articles explore how these languages transform software design approaches. I'm passionate about creating thoughtfully crafted systems with long-term maintainability.

Andres Monge

Experience Highlights - CTO, Software Architect, and Mentor

Executive and hands-on leadership across architecture, delivery, and platform reliability. These are the core roles that define my recent trajectory:

CTO - Technology Director @ Ultrai
Lead architecture and engineering execution for a financial services startup while remaining deeply hands-on (70% coding/spikes, 30% leadership).

Senior Python Software Engineer @ Origen AI
Led platform evolution and productionization of AI services, including compliance-critical delivery and resilient workflow architecture.

Engineering Manager @ Lavanda
Led two squads through delivery and technical debt reduction, balancing execution quality with business outcomes.

Technical Lead @ Santander Global Tech
Led international squads and large-scale banking modernization with API-first and event-driven patterns.

Explore detailed project implementations and technical achievements in the experience sections below.

Work Style - Artisan and INTP-A Logician

Code Quality Champion
Prioritize clean, maintainable code through rigorous code reviews and documentation standards, ensuring long-term project sustainability.

Iterative Architect
Balance immediate business needs with long-term technical vision through phased implementation strategies and evolutionary design patterns.

Documentation Advocate
Treat technical documentation as first-class engineering artifacts, creating living documents that streamline onboarding and knowledge sharing.

Cross-Disciplinary Engineer
Implement solutions that bridge technical domains by leveraging expertise in databases, infrastructure, and application development for cohesive systems.

CTO - Technology Director **Ultrai (Apr 2025-Present)**

Software Architecture, Monolith Design, Functional Programming Principles, Python, Rust, PostgreSQL, GitLab, Infisical, Authelia, NGINX, VictoriaLogs, Mattermost, Self-hosted Infrastructure, OWASP

At Ultrai, I lead technology execution end-to-end for a financial services startup while staying deeply hands-on (70% coding/spikes, 30% leadership). I own architecture, delivery, hiring, infrastructure, and security while co-driving roadmap and sprint planning.

- Built a new chatbot platform from scratch and enabled a product pivot to a dashboard-centric offering
- Designed a flexible monolith architecture that supported fast iteration without sacrificing stability or testability
- Introduced Rust for performance-critical components and enforced strict typing with stateless design practices
- Implemented secure PostgreSQL patterns with security definer functions to prevent unrestricted SQL access from the application layer
- Established and operated a fully self-hosted open-source stack (GitLab, Infisical, Authelia, NGINX, VictoriaLogs, Mattermost) with OWASP-aligned security practices

Principal Python Engineer **Origen AI (2024-2025)**

Python, Kubernetes, Docker, FastAPI, PostgreSQL, CI/CD, Security Compliance, Mentoring

As backend development lead for the Platform team, I guided the evolution of Proteus, Origen's client-facing AI platform. My days involved collaborating with data scientists to translate AI models into production-ready services, architecting workflow systems that balanced performance with reliability, and ensuring our multi-tenant infrastructure maintained strict client isolation.

- Secured SOC2 compliance certification for the core platform
- Upgraded AI platform from Python 3.7 to 3.12
- Developed BreezeFlow, a lightweight Airflow for AI pod management
- Implemented complete client isolation using Kubernetes namespaces

Personal Growth:

- Mastered sophisticated Kubernetes patterns specific to AI workloads
- Developed fluency in translating between data science and engineering terminology

Using Python and **FastAPI**, I proudly engineered testing protocols that decreased deployment failures by 65% while developing **fluency** in translating **between data science and engineering** terminology.

Senior Python Engineer **Babio (contractor) (2023-2024)**

Python, FastAPI, Pydantic, MongoDB, Docker, Cross-functional Leadership, Technical Mentoring

As project lead for Babio's solar proposal system, I architected and developed software that transformed the company's proposal generation process. Each day involved integrating complex data sources into a cohesive system that automatically generated tailored installation plans.

- Built complete system architecture using Python, MongoDB, and Docker
- Reduced proposal generation time by 70% through automated processing
- Achieved 90% test coverage ensuring exceptional system reliability

Personal Growth:

- Mastered modern Python web frameworks and API design patterns
- Solidified my test driven development, making my code 100% covered

Architecting with **Python (FastAPI)**, I created a system so robust it became the unchanged **core of Babio's business** while reducing proposal generation time by 70%

Machine Learning Engineer **Delorian Group (contractor) (2023-2023)**

Python, PyTorch, Transformers, CNN Architecture, Data Preprocessing, Independent Project Management

As the sole developer for this three-month AI project, I independently built a landmark recognition system from the ground up. Managed the entire ML lifecycle while ensuring seamless integration with existing tourism systems.

- Developed CNN model with 83% accuracy in landmark identification
- Delivered complete end-to-end solution enhancing tourism offerings

Personal Growth:

- Expanded Python expertise into specialized AI/ML applications
- Developed deeper understanding of ML infrastructure requirements

Leveraging PyTorch, I independently developed a **CNN model** achieving 83% landmark identification accuracy while expanding my specialized AI/ML expertise.

Engineering Manager **Lavanda (2022-2023)**

Agile/Scrum methodologies, Jira, Git workflow management, Team Leadership, Strategic Planning

Led two cross-functional engineering squads (10 developers) through product evolution phases, focusing on customer retention initiatives and technical debt management while bridging technical and business priorities.

- Reduced customer churn by 18% through strategic feature implementation
- Decreased sprint planning time by 40% through process refinements

Personal Growth:

- Developed nuanced perspective on balancing technical excellence with business outcomes
- Refined approach to nurturing technical talent through individualized plans

Through strategic planning, I refined approaches to nurturing technical talent while creating documentation systems adopted company-wide.

Mentor & Frontend Engineer

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Technical Lead **Global Savings Group (2021-2022)**

JavaScript (ES6+), NodeJS, Python, Docker, Functional Programming, Team Leadership, Legacy Modernization

Guided cross-functional team through legacy system modernization while maintaining critical banking services. Established technical standards and code review processes ensuring stability during transitions.

- Designed NodeJS web-crawler saving €500/week through FP principles
- Reduced system downtime 35% via proactive monitoring solutions
- Established coding standards cutting bug reports by 30%

Personal Growth:

- Mastered FP principles for efficient data processing
- Enhanced NodeJS expertise through pure functional implementations
- Developed techniques for technical debt/business balance

Applying NodeJS with **functional programming principles**, I designed a web-crawler saving €500/week while mastering FP techniques for efficient data processing.

Technical Lead **Santander Global Tech (2018-2021)**

Angular, Python, Event-Driven Architecture, MuleSoft, API-first Design, International Leadership

Spearheaded modernization of banking backend services across Spain/US teams. Established MuleSoft and API-first patterns while ensuring compliance.

- Led 10 international engineering squads as promoted Technical Lead
- Designed MyPage system & enterprise component library (10+ teams)
- Developed Python automation tools improving delivery predictability 30%

Personal Growth:

- Mastered large-scale distributed systems design
- Enhanced API-first/event-driven architecture expertise
- Strengthened international team leadership

Harnessing Python, I developed **automation tools** that improved delivery predictability by 30% while mastering large-scale distributed systems design

Principal Frontend Engineer **Bonial / Kaufda.de (2016-2017)**

Polymer, Micro-frontends, AI Training Systems, UI/UX Optimization, Performance Engineering

Mentored front-end team developing micro-frontends with Polymer for AI training systems while focusing on high-performance UI components and optimized user experiences.

- Created an infinite undo/redo global interface enhancing user workflow
- Engineered hyper-fast UI tag autocomplete utilizing local storage optimization
- Implemented grid splitter directly into AI tool, accelerating section selection by 300x

Personal Growth:

- Gained valuable expertise in AI training systems
- Developed advanced proficiency in Polymer framework
- Mastered performance optimization techniques for complex UIs

Pioneering UI performance, I engineered a **grid selection system** that accelerated AI training workflows by 300x while delivering exceptional user experiences

Principal Frontend Engineer **BBVA (2013-2017)**

AngularJS, Polymer.js, ReactJS, Cordova, Component Architecture, Cross-platform Dev

Directed frontend development for BBVA banking projects, leading R&D efforts and technological transitions with banking-grade security.

- Co-designed TabletBanking app (BBVA global standard)
- Mentored 3 junior developers to career advancement
- Created 200+ Polymer.js components using TDD

Personal Growth:

- Mastered cross-platform enterprise architecture
- Implemented TDD methodologies

Using **Polymer.js**, I **co-designed** the TabletBanking app that became **BBVA's global standard** while mastering cross-platform enterprise architecture

Frontend Engineer — GFT (2012-2013)

NodeJS, AngularJS, CSS, Bash/Python, Component Development

Developed enterprise frontend solutions for insurance sector, mentoring teams through jQuery to Angular transition.

- Built personal finance manager in 6 weeks
- Created Mapfre Component Catalogue
- Developed Angular/ReactJS hybrid component

Personal Growth:

- Refined mentoring through framework transitions
- Deepened component architecture expertise
- Strengthened reusable component development

Leveraging **Angular**, I **created the Mapfre Component Catalogue** while deepening my expertise in component architecture and reusability.

Web Developer — C.B.I. (2010-2012)

SQL, PHP, JavaScript, WordPress, jQuery, Database Optimization, Enterprise Applications

Focused on backend development using SQL functions with PHP connecting database operations to JavaScript frontends. Contributed to enterprise CRM-like application supporting core business operations while maintaining proprietary frameworks.

- Developed components for enterprise-grade application similar to SAP
- Reduced database query execution time by 35% through SQL optimization
- Implemented caching strategies improving response time by 45%
- Ensured cross-browser compatibility including legacy IE7 support

Personal Growth:

- Developed expertise in relational database operations
- Discovered functional programming concepts through SQL
- Built foundation for backend specialization

Applying **SQL functions** with functional programming principles, I **reduced database query** execution time by 35% while developing expertise in relational database operations.

Web Developer — Freelance (2008-2013)

PHP, Vanilla JavaScript, Joomla, WordPress, Drupal, Full-stack Development, Client Management

Designed custom websites for international clients, managing full development lifecycle from consultation to deployment. Created optimized themes/plugins and maintained cross-platform compatibility.

- Developed Universidad Rey Juan Carlos website (www.urjc.es)
- Created Los Hijos de la Malinche cultural platform
- Delivered 10+ websites with 100% client satisfaction
- Improved site loading times by 40% through theme optimization

Personal Growth:

- Gained holistic perspective on web application lifecycles
- Enhanced client communication with diverse stakeholders
- Strengthened independent problem-solving abilities

Using **WordPress**, I developed the Universidad Rey Juan Carlos website while **gaining a holistic perspective** on web application lifecycles.

Web Developer Intern **Know-Century (2007-2008)**

HTML, PHP, CSS, JavaScript, CMS, SEO

Initiated as web developer before establishing company's SEO department. Optimized digital assets and coordinated technical/content teams for search performance.

- Created first formal SEO workflow and optimization process
- Achieved 2nd rank for "comprar oro" (competitive keyword)
- Developed comprehensive SEO style guide for content teams

Personal Growth:

- Discovered passion for technical problem-solving
- Understood frontend-data structure relationships
- Sparked interest in system architecture

Through **engineering approaches and creative optimization**, I achieved **2nd rank** for "comprar oro" while discovering my passion for technical problem-solving.